

Lab Department Implementation

Frontend Developer Guide

Date: 11 August 2026 | Audience: Frontend developers | Scope: Laboratory vs Radiology departments for lab orders and lab technicians.

IMPORTANT

Backend for department routing is ALREADY DONE. This document describes frontend-only updates. Do not change backend files for this feature.

1. Product Goal

The Lab module has two departments. Lab technicians belong to one department and only see orders for that department. Doctors choose the lab department when ordering a test, then only see tests for that department.

Department	Code	Example tests
Laboratory (Pathology)	LAB	Blood, Urine, CBC, Biochemistry
Radiology (Imaging)	RAD	X-Ray, USG, CT, MRI, Mammography

Where Laboratory / Radiology MUST appear

- Doctor module - when creating a lab order (department picker + filtered tests)
- Admin / Super-admin - when creating or editing a lab technician (assign LAB or RAD)

Where Laboratory must NOT appear

- OPD Billing / Register Patient / Book Appointment - clinical doctor department dropdown
- That dropdown is for Cardiology, Pediatrics, etc. Lab departments are not visit departments.

2. End-to-End Flow

A. Admin assigns lab tech department

- Create or edit staff with role lab_technician
- Require department: Laboratory (LAB) or Radiology (RAD) only
- POST /register or PATCH staff update with department_id

B. Doctor orders a lab test

- Open consultation / Labs section
- Select Lab Department: Laboratory OR Radiology
- Test dropdown shows only that department's tests
- POST /lab-tests with department_id (+ test_name, category, priority, ...)

C. Lab tech login

- Same login as today; role remains lab_technician
- GET /lab/orders, /lab/dashboard, /lab/reports are auto-filtered by backend
- Tech with no department_id gets HTTP 403 until admin assigns LAB/RAD

3. Backend Contract (Already Implemented)

Frontend must send and display department_id. Backend enforces scoping.

3.1 Create lab order - POST /lab-tests

```
{
  "appointment_id": 123,
  "test_name": "CBC",
  "category": "Blood",
  "department_id": 11,
  "priority": "Normal",
  "clinical_notes": "optional"
}
```

- department_id is optional for backward compatibility
- If omitted, backend INFERS: Radiology-like category/test_name -> RAD, else -> LAB
- Prefer ALWAYS sending department_id from the new doctor UI

3.2 Lab order responses

- LabTestResponse / list items include department_id
- Lab technician order list/detail also include department_id

3.3 Lab technician APIs (auto-scoped)

- GET /lab/dashboard
- GET /lab/orders and GET /lab/orders/{id}
- GET /lab/reports and report detail/file
- PATCH sample-collected / complete, POST report / upload-file
- All filtered or blocked by current_user.department_id

3.4 Staff register / update

- Lab technician may store department_id (must be LAB or RAD code)
- Doctor still uses any clinical department
- OPD nurse etc. still do not use department_id on user

4. Frontend Files To Update

NO BACKEND CHANGES

Update only HM-frontend-Side files listed below.

4.1 Hide LAB from OPD clinical department dropdown

File	What to do
src/shared/api/services/opdReference.js	Filter out code LAB (and RAD if needed)

Best place: listDepartments(). Filter departments where code is LAB (and optionally RAD) so Register Patient, OPD Bill, Book Appointment, etc. all stop showing Laboratory without editing every page.

```
// Example in listDepartments
const LAB_ONLY_CODES = new Set(['LAB']); // or ['LAB', 'RAD']
return rows
  .map(apiToUiDepartment)
  .filter(Boolean)
  .filter((d) => !LAB_ONLY_CODES.has(String(d.code || '').toUpperCase()));
```

4.2 Doctor lab order UI

File	What to do
src/features/doctor/constants.js	Add LAB/RAD departments + tests per dept
.../doctor/components/ConsultationModal.jsx	Dept select -> filter tests
.../doctor/components/LabsSection.jsx	Same for create/edit lab order
src/shared/api/mappers/clinicalMapper.js	Send department_id in create/update
src/shared/api/services/doctorLabs.js	Only if payload wiring needs tweak

4.3 Admin assign department to lab tech

File	What to do
features/admin/pages/StaffRegisterPage.jsx	Dept required for lab_technician
features/admin/pages/StaffDetailPage.jsx	roleRequiresDepartment + LAB/RAD only
features/super-admin/...StaffRegisterPage.jsx	Same if used
features/super-admin/...StaffDetailPage.jsx	Same if used

5. Doctor Module - Detailed UI Spec

5.1 Suggested constants.js shape

```
export const LAB_DEPARTMENTS = [
  { code: 'LAB', label: 'Laboratory' },
  { code: 'RAD', label: 'Radiology' },
];

export const LAB_TESTS_BY_DEPARTMENT = {
  LAB: [
    'Blood Test', 'Urine Test', 'Stool Test',
    'Biochemistry', 'Hematology', 'Microbiology',
    'Histopathology', 'CBC', 'Lipid Profile',
    'Blood Sugar', 'Urine Routine',
  ],
  RAD: [
    'X-Ray', 'Ultrasound (USG)', 'CT Scan',
    'MRI', 'Mammography',
    'X-Ray Chest', 'MRI Brain', 'CT Scan Abdomen',
  ],
};

// Keep category as a soft label if needed (Blood/Urine/etc.),
// but department is the routing field.
```

5.2 Consultation / Labs UX rules

- Show Lab Department dropdown FIRST (Laboratory | Radiology)
- Load department options from /departments or /opd/departments filtered to LAB+RAD
- OR hardcode labels and resolve id by matching department.code
- Test dropdown options = LAB_TESTS_BY_DEPARTMENT[selectedCode]
- If department changes, CLEAR selected test
- Disable test dropdown until department is selected
- On save, send department_id (numeric id from departments API)
- Also send category for display (e.g. Blood / Radiology) if UI still uses it

5.3 clinicalMapper.js - create payload

```
export function uiToApiLabTestCreate(ui) {
  return {
    appointment_id: ui.appointmentDbId,
    test_name: ui.testName ?? ui.test,
    category: ui.category,
    department_id: ui.departmentId ?? ui.department_id, // REQUIRED from new UI
    priority: ui.priority || 'Normal',
    clinical_notes: ui.clinicalNotes ?? ui.clinical_notes ?? '',
  };
}

// Also map department_id on update if doctor can change department/test.
```

5.4 Display department on doctor Labs list (optional but useful)

- apiToUiLabTest: map department_id / departmentName if present

- Show a badge: Laboratory or Radiology next to the order

6. Admin / Super-Admin - Lab Tech Department

Today department is required only for doctor. Extend the same pattern to lab_technician, but limit options to LAB and RAD.

- needsDepartment / roleRequiresDepartment: doctor OR lab_technician
- Department options for lab_technician: filter departments where code is LAB or RAD
- For doctor: keep full clinical department list (exclude LAB from doctor list too)
- Validation toast: 'Please select Laboratory or Radiology for lab technician'
- Submit department_id in register and update payloads

```
const isLabTech = selectedRole?.name === 'lab_technician';
const needsDepartment =
  selectedRole?.name === 'doctor' || isLabTech;

const departmentOptions = departments
  .filter((d) => {
    const code = String(d.code || '').toUpperCase();
    if (isLabTech) return code === 'LAB' || code === 'RAD';
    // doctor clinical list - hide lab-only depts
    return code !== 'LAB';
  })
  .map((d) => ({ value: String(d.id), label: d.name }));
```

7. Lab Technician Frontend

Most lab tech pages need little change because backend already scopes data. Recommended UX polish:

- Show current department name in LabLayout / LabAppShell header
- If API returns 403 (no department assigned), show clear message to contact admin
- Do not rely on client-side category filter for security - backend already filters
- Optional: map department_id in labMapper.js for display badges

Optional polish files	Action
features/lab/components/LabLayout.jsx	Show LAB/RAD label
features/lab/components/LabAppShell.jsx	Same
shared/api/mappers/labMapper.js	Map department_id for UI

8. How Frontend Gets department_id

Doctor UI needs numeric department_id for POST /lab-tests. Use existing departments API.

- Fetch departments list (admin /departments or OPD /opd/departments)
- Find row where code === 'LAB' or code === 'RAD'
- Use that row's id as department_id

```
function resolveLabDepartmentId(departments, code) {
  const row = departments.find(
    (d) => String(d.code || '').toUpperCase() === code
  );
  return row ? Number(row.id) : null;
}

// On save:
department_id: resolveLabDepartmentId(allDepartments, selectedCode) // 'LAB' | 'RAD'
```

SEED REQUIREMENT

Database must have departments Laboratory (LAB) and Radiology (RAD). Backend seed.py already includes both. Run seed if LAB is missing.

9. Acceptance Criteria (QA Checklist)

OPD Billing

- Register Patient department dropdown does NOT show Laboratory
- Doctor clinical departments still appear (Cardiology, Pediatrics, ...)

Doctor module

- Lab order form shows Laboratory / Radiology department selector
- Selecting Laboratory shows only laboratory tests
- Selecting Radiology shows only radiology/imaging tests
- Changing department clears previously selected test
- Created order includes department_id in network payload

Admin

- Creating lab_technician requires selecting Laboratory or Radiology
- Cannot assign Cardiology etc. to lab technician

Lab technician

- Laboratory tech sees only Laboratory orders
- Radiology tech sees only Radiology orders
- Cannot open other department order by URL/id (404/403 from API)
- Notifications only for own department orders

10. Out of Scope

- No new lab_test_catalog table / model
- No new role names (keep single role: lab_technician)
- No new /lab routes - same portal, filtered data
- No backend file changes for this frontend task

11. Suggested Frontend Implementation Order

- 1) opdReference.js - hide LAB from OPD clinical dropdowns (quick win)
- 2) Admin staff register/detail - assign LAB/RAD to lab techs
- 3) doctor/constants.js - department + test maps
- 4) clinicalMapper.js - add department_id to create/update
- 5) ConsultationModal.jsx + LabsSection.jsx - UI wiring
- 6) Optional lab shell badge + 403 empty state

Document end - Frontend Lab Department Guide